OMARI SEMINAR SERIES

Beyond the AUC

Developing and validating clinical prediction models

A case study in metabolic gestational age estimation

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QUICK POLL · HANDS UP

A prediction model has an AUC of 0.95. Would you use it in your clinic?

A
Yes, that's excellent

B
No

C
It depends…

A good AUC can hide a bad model

1

Wrong for individuals

Ranks patients well, but the predicted values are off.

2

Fails when it travels

Works in the development data, breaks in a new lab or population.

3

Doesn't help decisions

Statistically 'good', clinically no better than treating everyone.

Today: how to develop and evaluate a model honestly, and how to read the performance numbers in papers.

Roadmap for the next two hours

5:00	Start with the question: what is the model for?
5:10	Accuracy for continuous predictions
5:30	Discrimination: the AUC and what it hides
5:45	Calibration: is 20% really 20%?
6:00	*Short break*
6:05	Clinical usefulness: net benefit and decision curves
6:15	When models travel: validation, shift, updating, pipeline pitfalls
6:40	What we'd do differently · discussion · wrap-up

The case study: gestational age from a heel-prick blood spot

- Gestational age (GA) drives newborn care and preterm birth statistics, but accurate dating ultrasound is often unavailable in low-resource settings
- Routine newborn screening already measures dozens of metabolites in a dried blood spot
- Machine learning models developed on Ontario newborn screening data estimate GA from clinical data plus these analytes
- Externally validated in Bangladesh, Zambia, Kenya, China and the Philippines
- Today we use the real models and open data to show why performance changes when a model travels, with a candid look at what we'd do differently

Companion materials

Part 1 · simulated data
demo.qmd: every metric in this talk, in base R

Part 2 · real, open data
demo_real.qmd: Zambia and Bangladesh cohorts (Dryad, doi:10.5061/dryad.m37pvmd6b) scored with the fitted Ontario models

PART 1 · 5:00

Start with the question

What the model is for decides how to build it and how to judge it.

Same model, three jobs, three report cards

Estimate one infant's GA

Used for

Clinical care, eligibility for interventions

Judge it by

Error in weeks (MAE), % within ±1 and ±2 weeks, bias by subgroup

Flag infants who were born preterm

Used for

Follow-up, referral

Judge it by

Sensitivity, PPV at a threshold, calibration of risk, net benefit

Estimate the preterm birth rate

Used for

Population surveillance

Judge it by

Calibration of the average: does the predicted rate match the true rate?

Ask 'what is it for?' before you look at a single metric.

The simulated data behind Part 1

Each infant has a true GA and a reference GA (what a dating ultrasound records). Real studies only see the reference.

	Development	External site A	External site B
Population	Reference population	Smaller for GA, more preterm	Smaller for GA, more preterm
Laboratory	Reference lab	Different calibration, 6 analytes	Different calibration, 5 analytes
Reference standard	First-trimester ultrasound	First-trimester ultrasound	Mostly 2nd/3rd-trimester ultrasound

- Clinical-only model: sex, multiple birth, birthweight (natural spline), OLS
- Full model: clinical + 30 log-analytes + squares + age at collection, elastic net ($\alpha = 0.5$, λ by 10-fold CV)
- Comparators: a random forest, and a deliberately overfitted model fitted on only 200 infants

PART 2 · 5:10

Accuracy for continuous predictions

Error metrics, calibration plots, regression to the mean, and subgroups.

Four numbers worth reporting for a continuous prediction

MAE

Typical error in weeks. 0.7 weeks is about ±5 days.

Bias

Mean predicted − reference: calibration-in-the-large for a continuous outcome.

% within 7 / 14 days

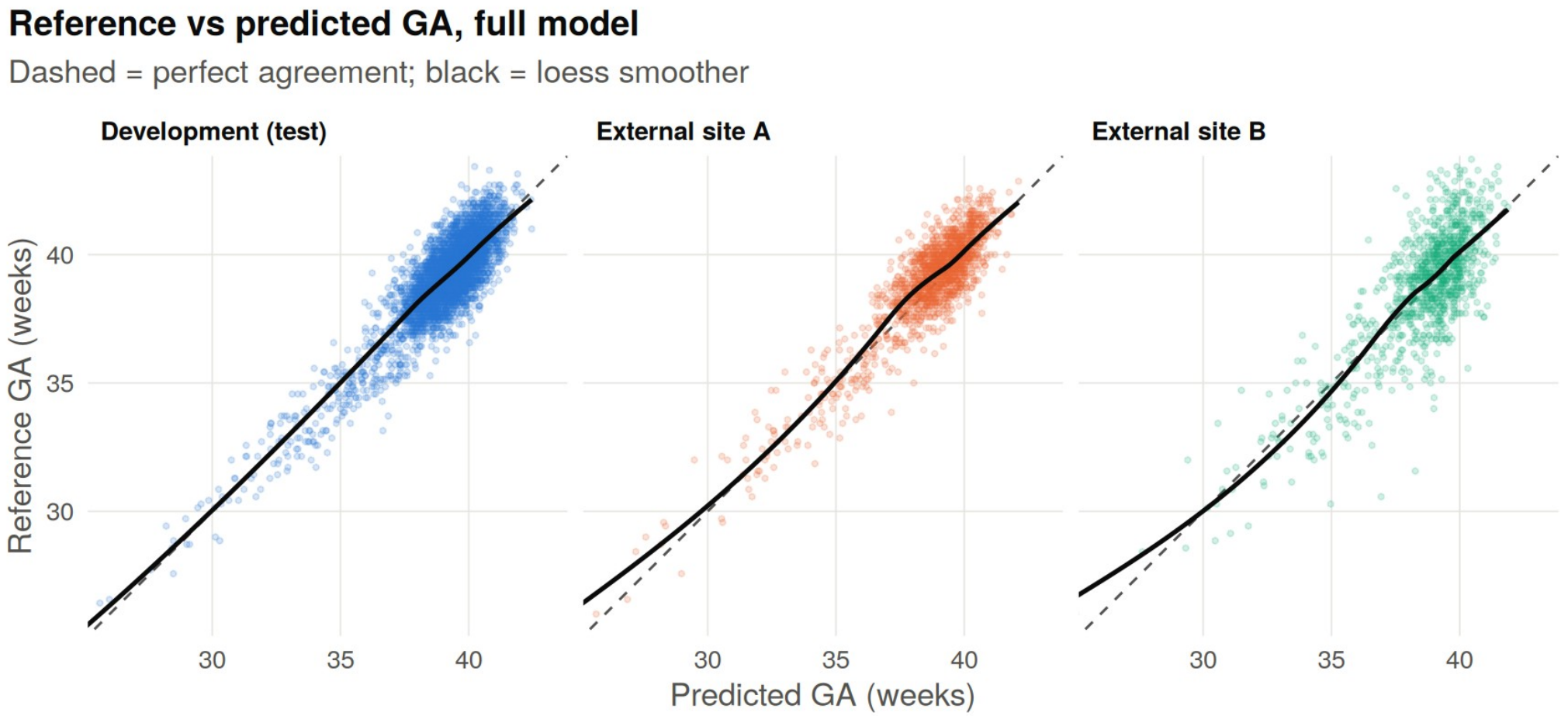
Often the most meaningful number for clinicians.

Calibration slope

Regress reference on predicted. Slope 1 = predictions neither too extreme nor too moderate.

R^2 and correlation alone don't tell a clinician how wrong a single prediction can be.

Reference vs predicted GA: looks great, right?



Dashed = perfect agreement; black = loess smoother. Watch the preterm end.

QUICK POLL · HANDS UP

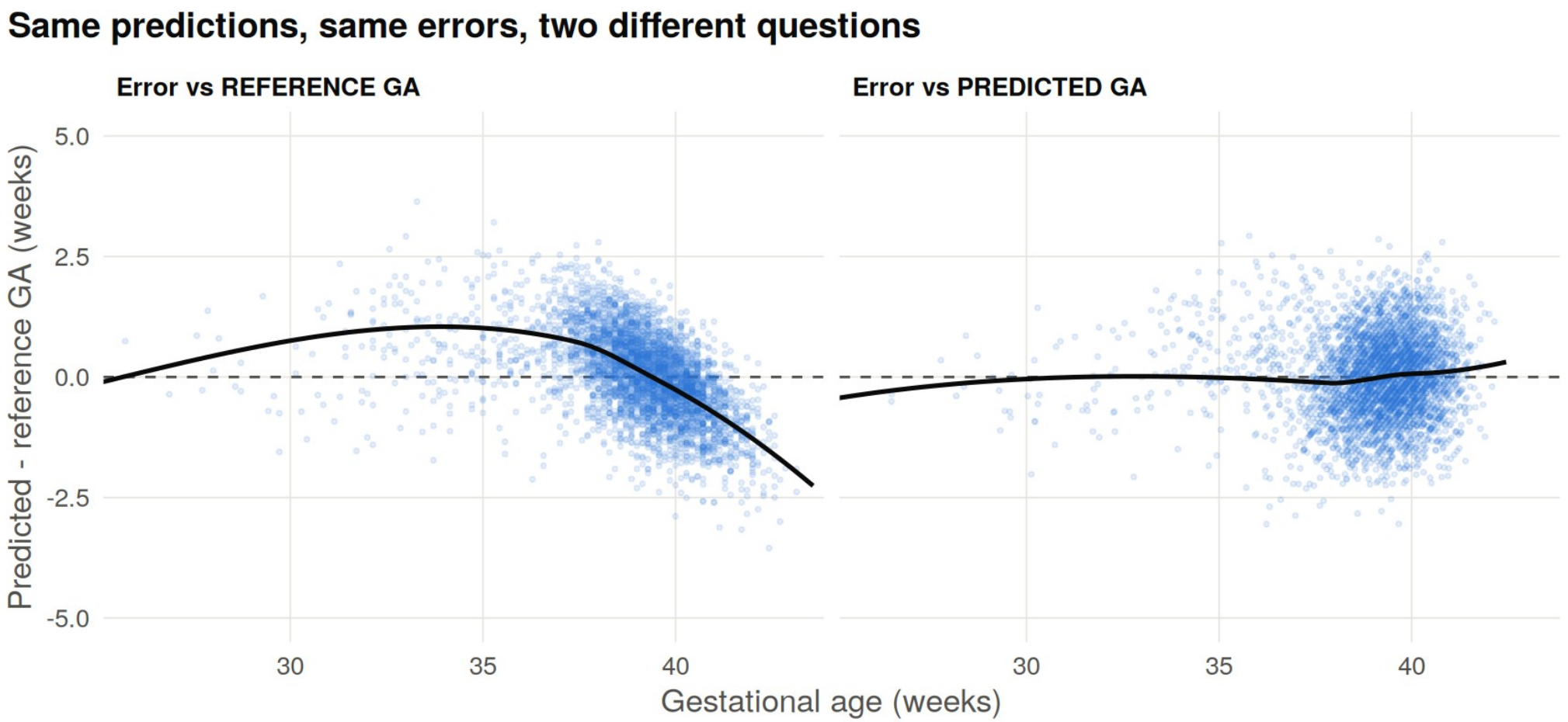
“The model overestimates GA in preterm infants.” Is the model miscalibrated?

A
Yes

B
No

C
Can't tell from that statement

Same predictions, same errors, two different questions



Grouped by true GA: regression to the mean. Grouped by predicted GA: broadly calibrated.

When the reference standard is the problem

- Site B looks much worse than site A
- Because the data are simulated, we can also score the predictions against the true GA
- Against the truth, the model is as accurate at site B as anywhere else
- Most of the apparent loss comes from a noisier reference standard (late ultrasounds), not the model

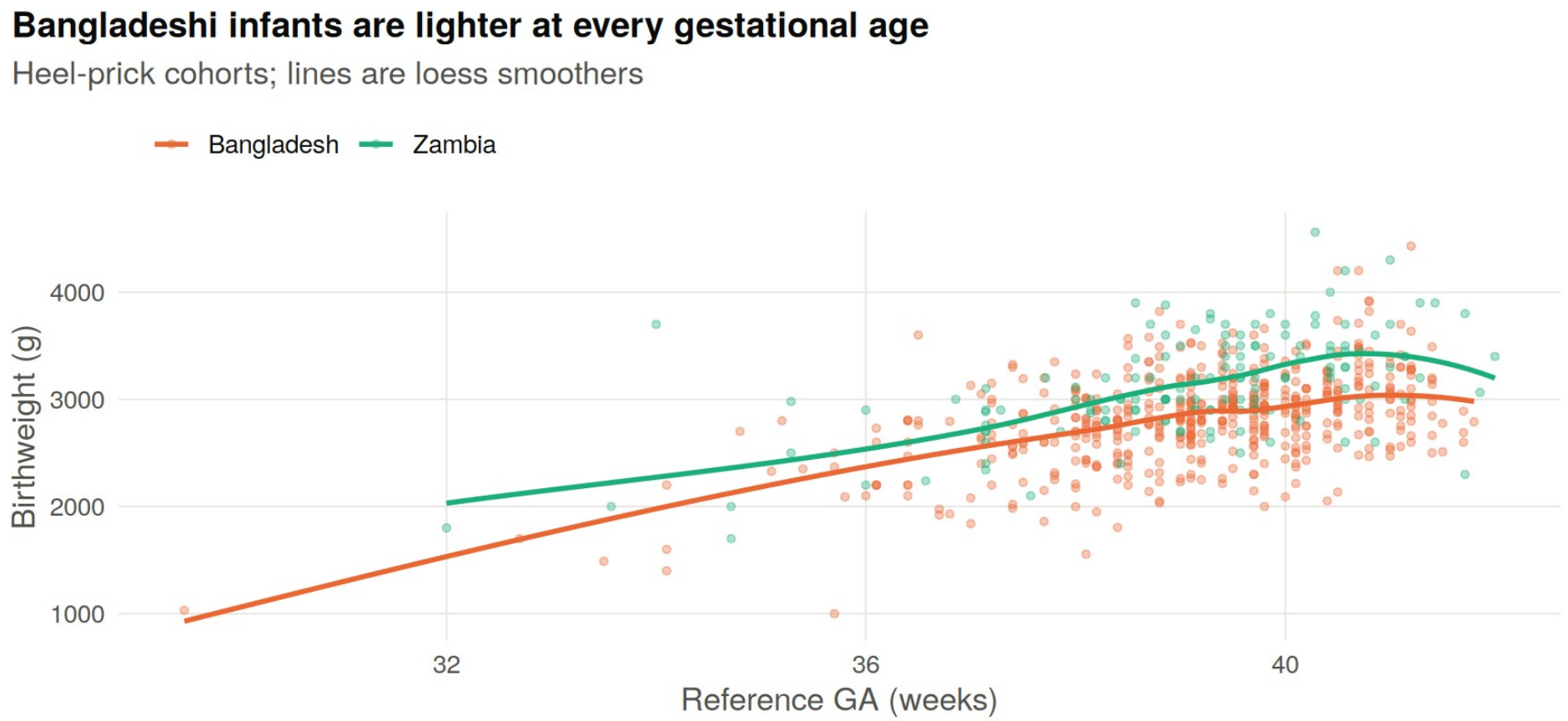
In real data you never see the truth

Check it indirectly: restrict to infants with the best-quality reference, as in the Kenya validation.

Interrogate the reference standard before blaming the model.

Real data: where the analytes earn their keep

Bangladeshi infants are lighter at every gestational age (heel-prick cohorts)



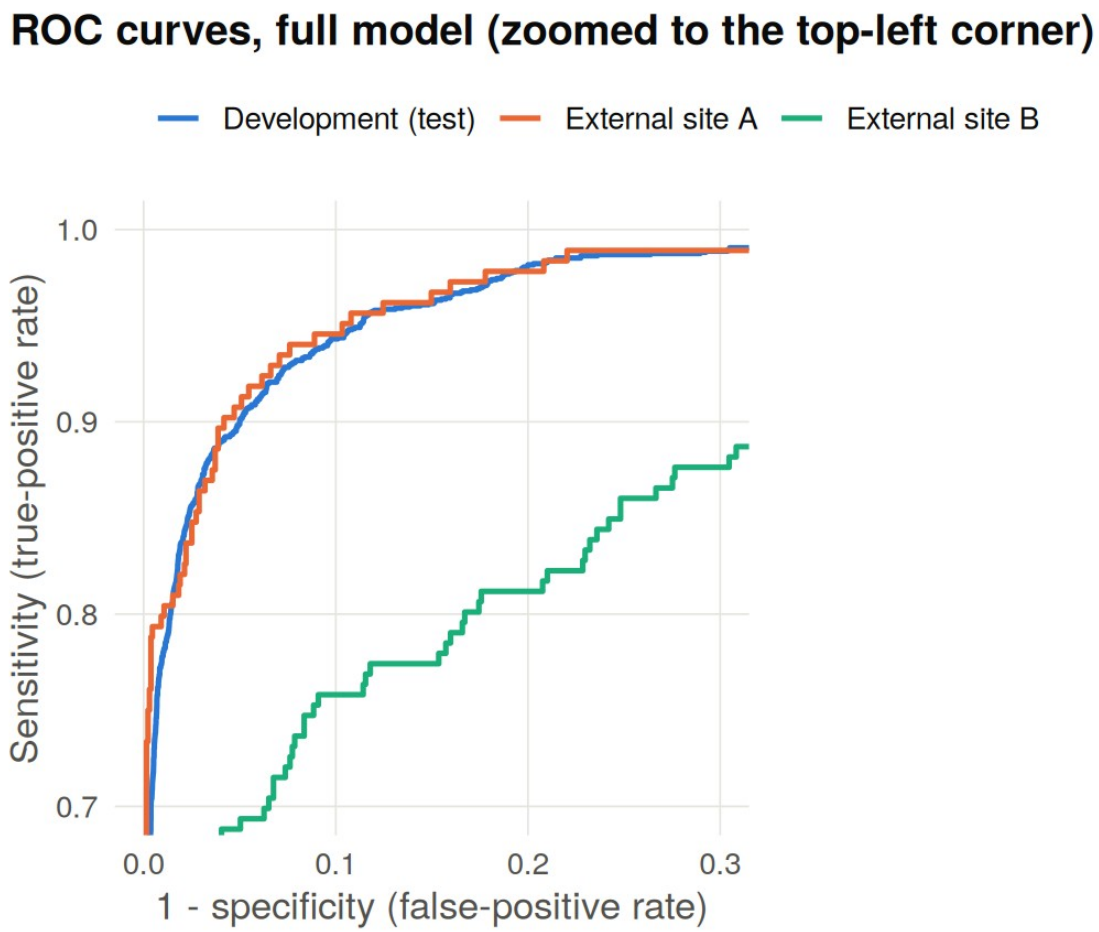
Birthweight models read growth restriction as immaturity: GA underestimated by 1.5–3.3 weeks in SGA infants.

PART 3 · 5:30

Discrimination

The AUC, what it hides, thresholds, and why rare outcomes kill 'accuracy'.

The AUC (c-statistic)



AUC = probability a random preterm infant gets a higher risk than a random term infant. Ranking only.

The accuracy trap

91.6%

accuracy of a 'model' that calls every infant term

- Accuracy is not a useful summary when outcomes are imbalanced
- Sensitivity and specificity describe the test
- PPV and NPV describe the patient in front of you, and they move with prevalence
- Raw Brier score depends on prevalence; use the scaled Brier (1 − Brier / Brier of the prevalence-only model) to compare sites

Real data: how sure are we about that AUC?

11

preterm infants in the Zambia heel cohort

35

in the Bangladesh heel cohort

0.64–0.99

95% CI for the birthweight model's AUC in Zambia heel samples

- Asymptotic intervals (e.g. DeLong) are themselves unreliable with 11 events
- Plan external validation sample size around the number of events (often 100 or more), and the precision needed for calibration as well as discrimination (Riley et al., BMJ 2024)

PART 4 · 5:45

Calibration

Is a predicted risk of 20% really 20%? Common failures and how to fix them.

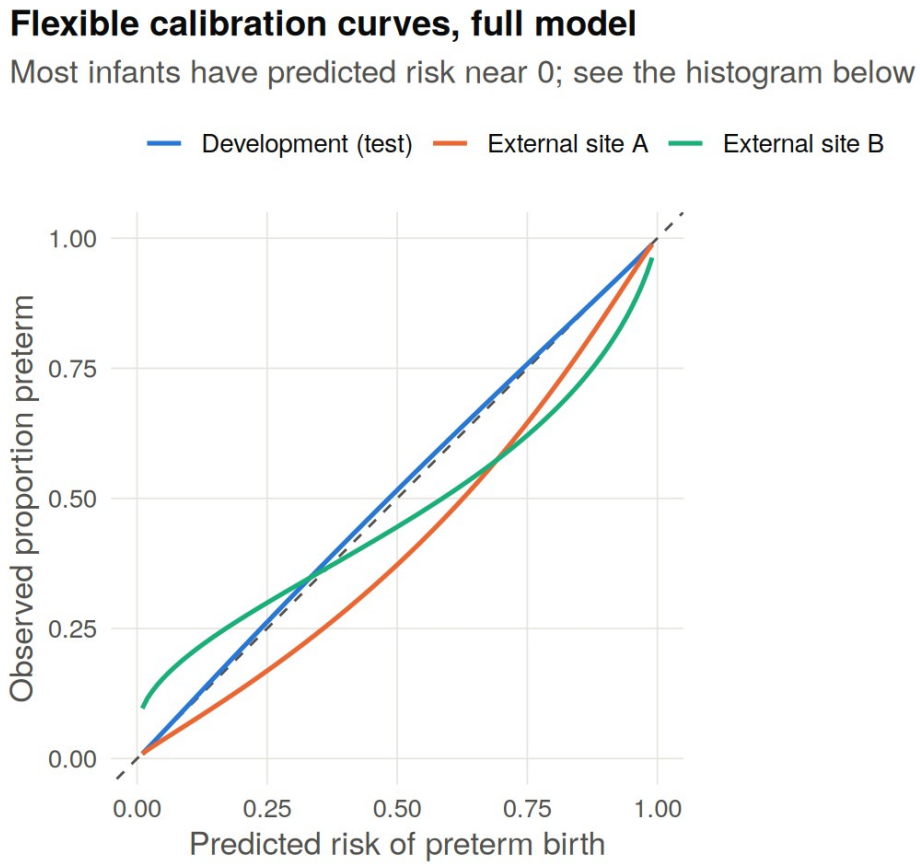
The calibration hierarchy

Level	Question	Measure
Mean	Is the average predicted risk right?	O/E ratio, calibration-in-the-large
Weak	Are predictions too extreme or too moderate?	Calibration slope
Moderate	Is predicted risk right across its whole range?	Flexible calibration curve

Report discrimination and calibration. Never the AUC alone.

Same AUC, different calibration

Flexible calibration curves, full model; most infants have predicted risk near 0



Site A ranks as well as development but over-predicts preterm risk. The AUC cannot see this.

Calibration slope: the overfitting detector

Slope < 1

Predictions too extreme

Small-n regression: 67 predictors, 200 infants. The classic sign of overfitting.

Slope > 1

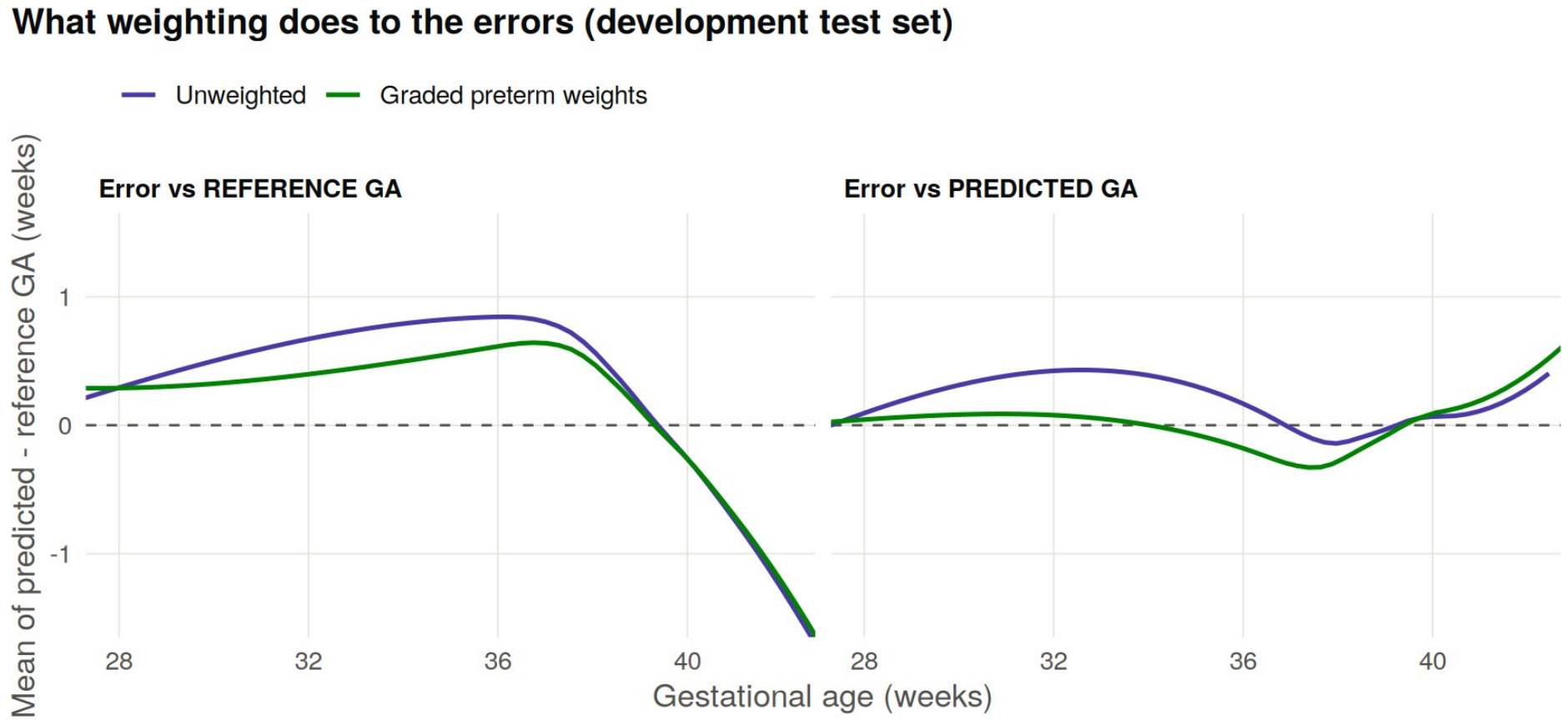
Predictions too compressed

Random forest: forests cannot extrapolate beyond the averages of their leaves.

Neither problem shows up in the MAE alone.

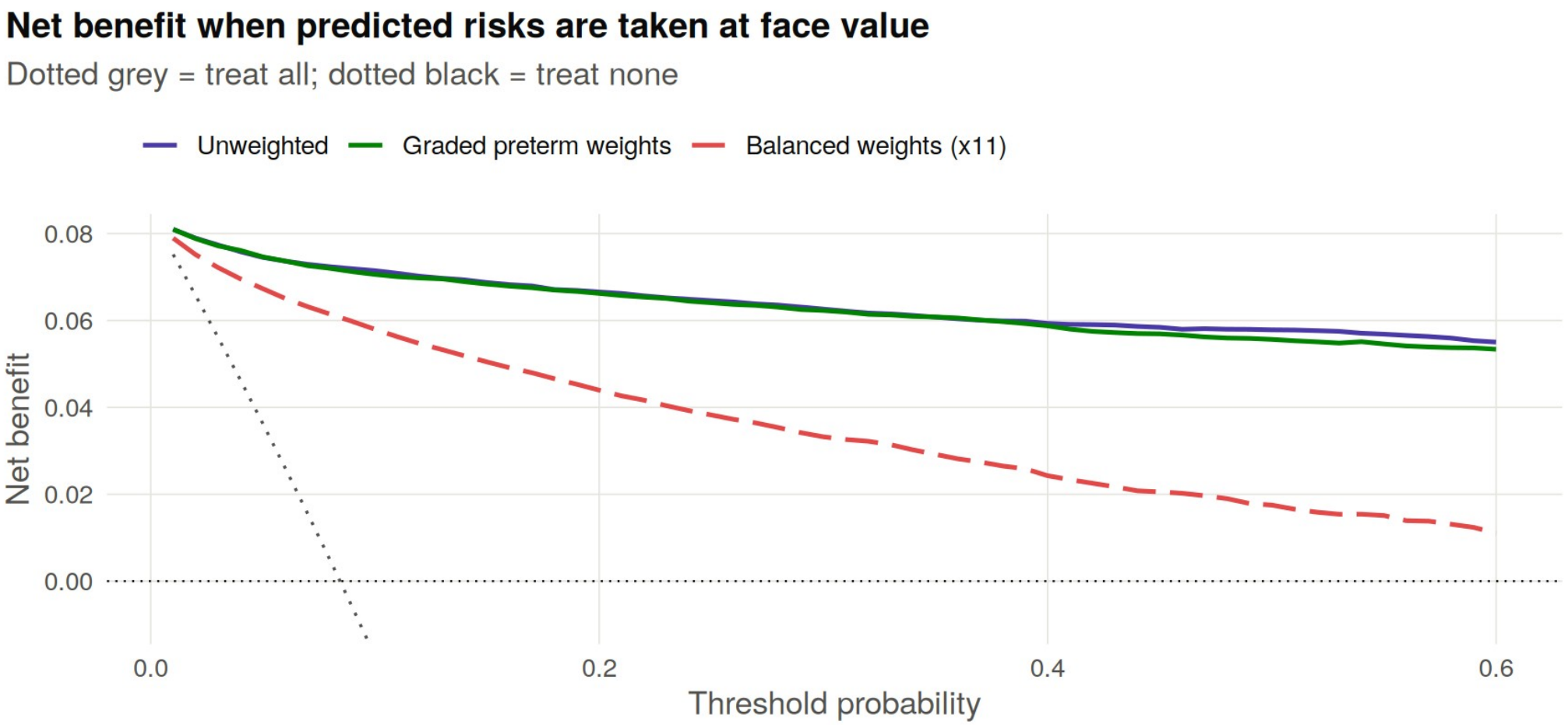
Class imbalance: should we upweight preterm infants?

Graded weights: w = 1 + (37 − GA)/1.5 below 37 weeks, so 34 weeks counts 3×, 28 weeks 7×



Weighting trades a little term accuracy for less preterm bias. It's a loss-function choice: report it as one.

Weighting moves the threshold, not the ranking



Balanced weights (×11) inflate every risk: same AUC, lower net benefit at face value.

Five-minute break

Question to ponder: which model in your own field is reported with an AUC and nothing else?

PART 5 · 6:05

Clinical usefulness

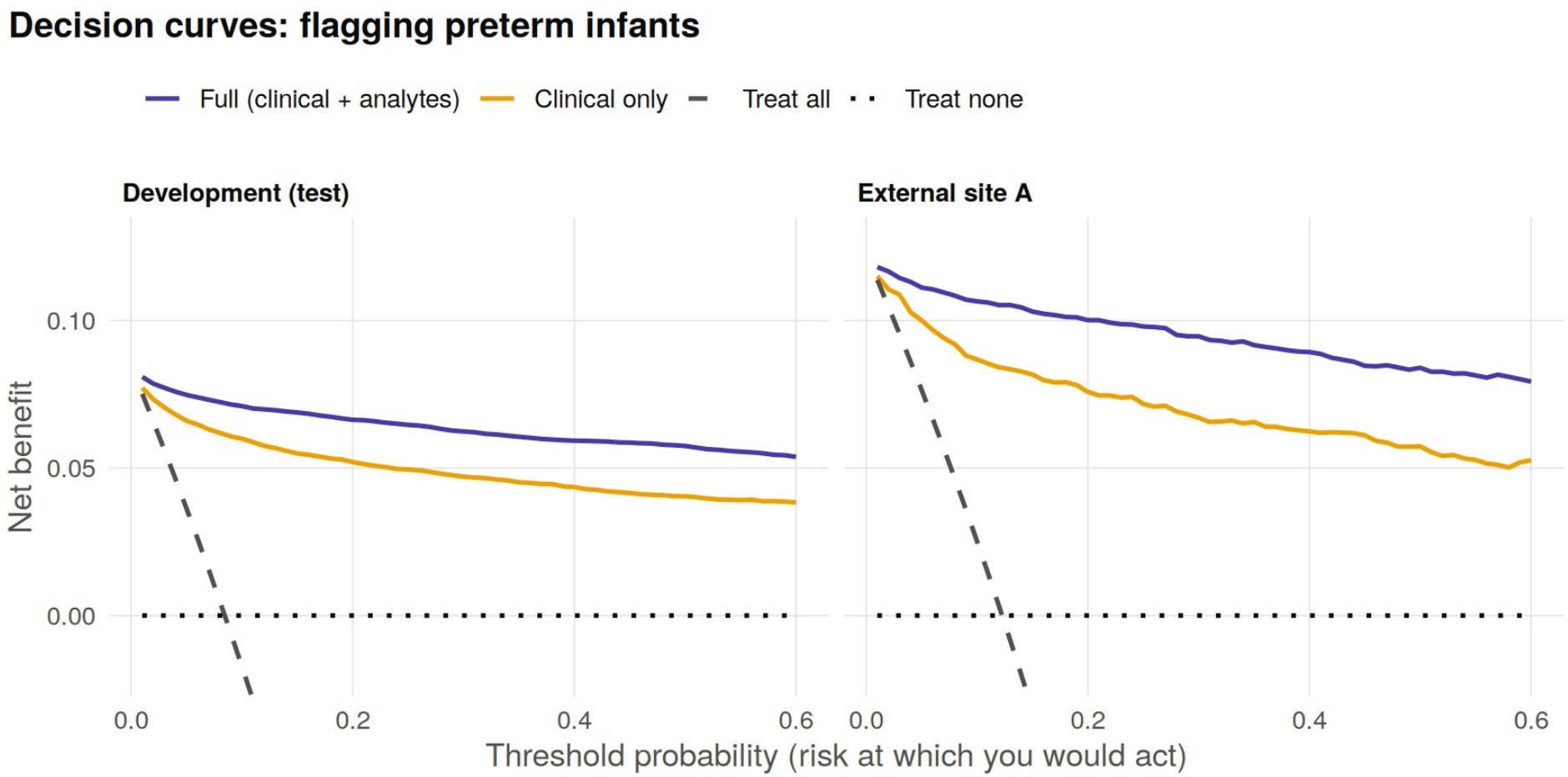
Net benefit and decision curves.

Net benefit puts a price on false positives

$$\text{Net benefit} = TP/n - FP/n \times p_t / (1 - p_t)$$

- The threshold probability p_t says how many false positives you'd accept per true positive
- $p_t = 0.2$ means flagging 4 term infants to catch 1 preterm infant is acceptable
- A model is useful at a threshold if it beats both 'treat all' and 'treat none'
- Choose the range of thresholds with clinicians, before looking at the results

Decision curves: flagging preterm infants



The full model adds net benefit over the clinical model across plausible thresholds, in both settings.

PART 6 · 6:15

When models travel

Internal vs external validation, shift, updating, and pitfalls hidden in the pipeline.

Internal and external validation answer different questions

Internal validation

How optimistic is the apparent performance?

Bootstrap optimism correction or cross-validation, using only the development data. Matters most when data are scarce.

External validation

Does it work somewhere else?

Tests transportability: new population, lab, sample type, reference standard. Plan the sample size around events.

The 200-infant model looks excellent on its own data. Bootstrap correction shows how much will disappear.

Four ways a new setting differs

Population

More SGA, more preterm (Bangladesh vs Zambia)

Sample type

Heel-prick vs cord blood in the same infants

Laboratory

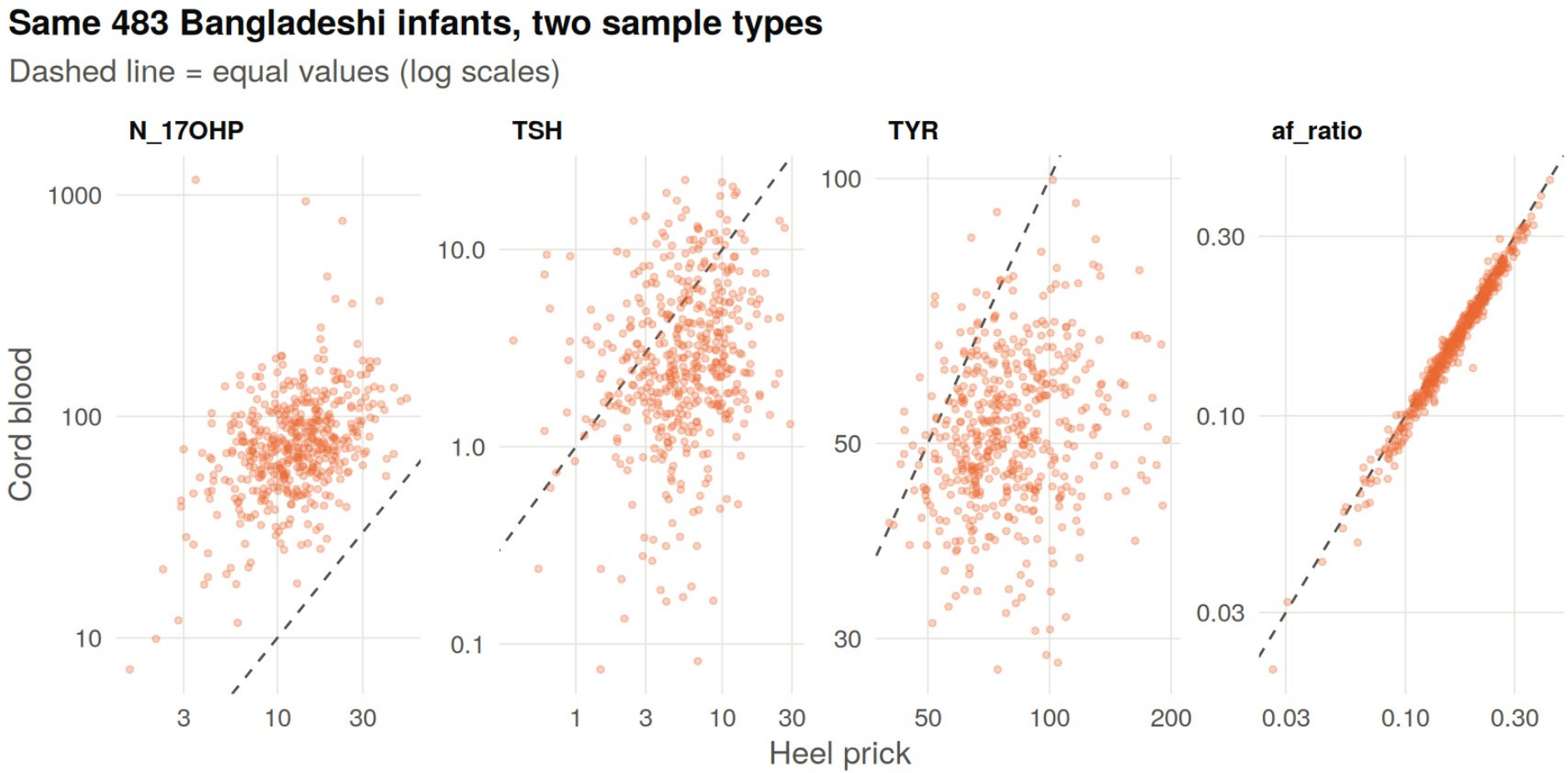
Different analyte calibration between labs

Reference standard

Late ultrasound vs first-trimester dating

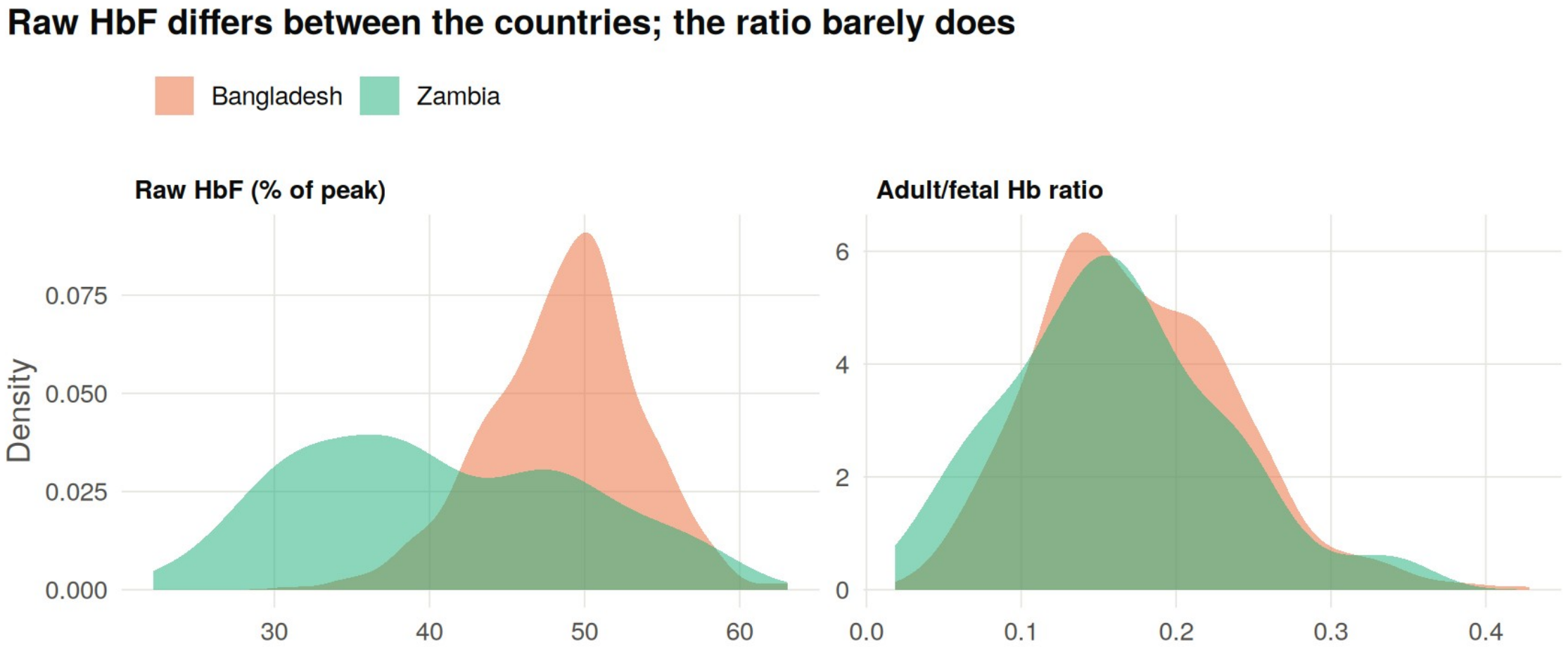
Look at covariate shift before you look at outcomes.

Heel and cord blood are different samples



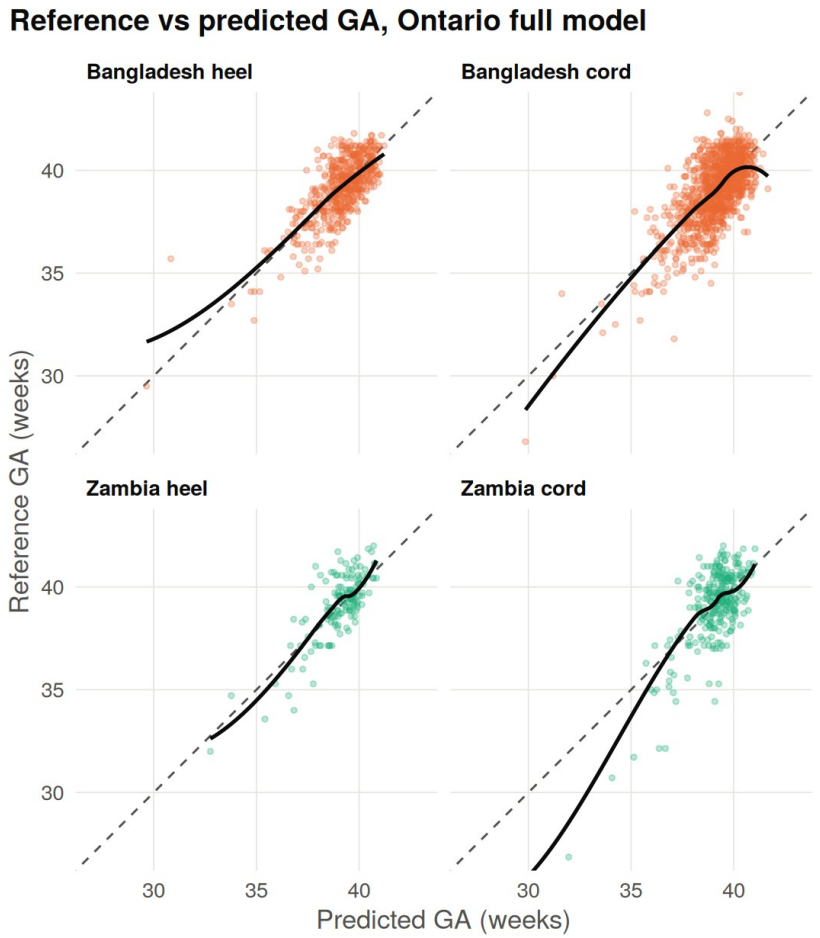
17-OHP is about 5× higher in cord blood. The same infant looks like a different population.

Labs differ; ratios travel better



A ratio of two measurements from the same run cancels much of the between-lab difference.

External validation of the Ontario model



Heel-prick MAE about 0.8 weeks in both countries, close to 0.7 in Ontario. Cord blood is worse.

Counting preterm births: a surveillance trap

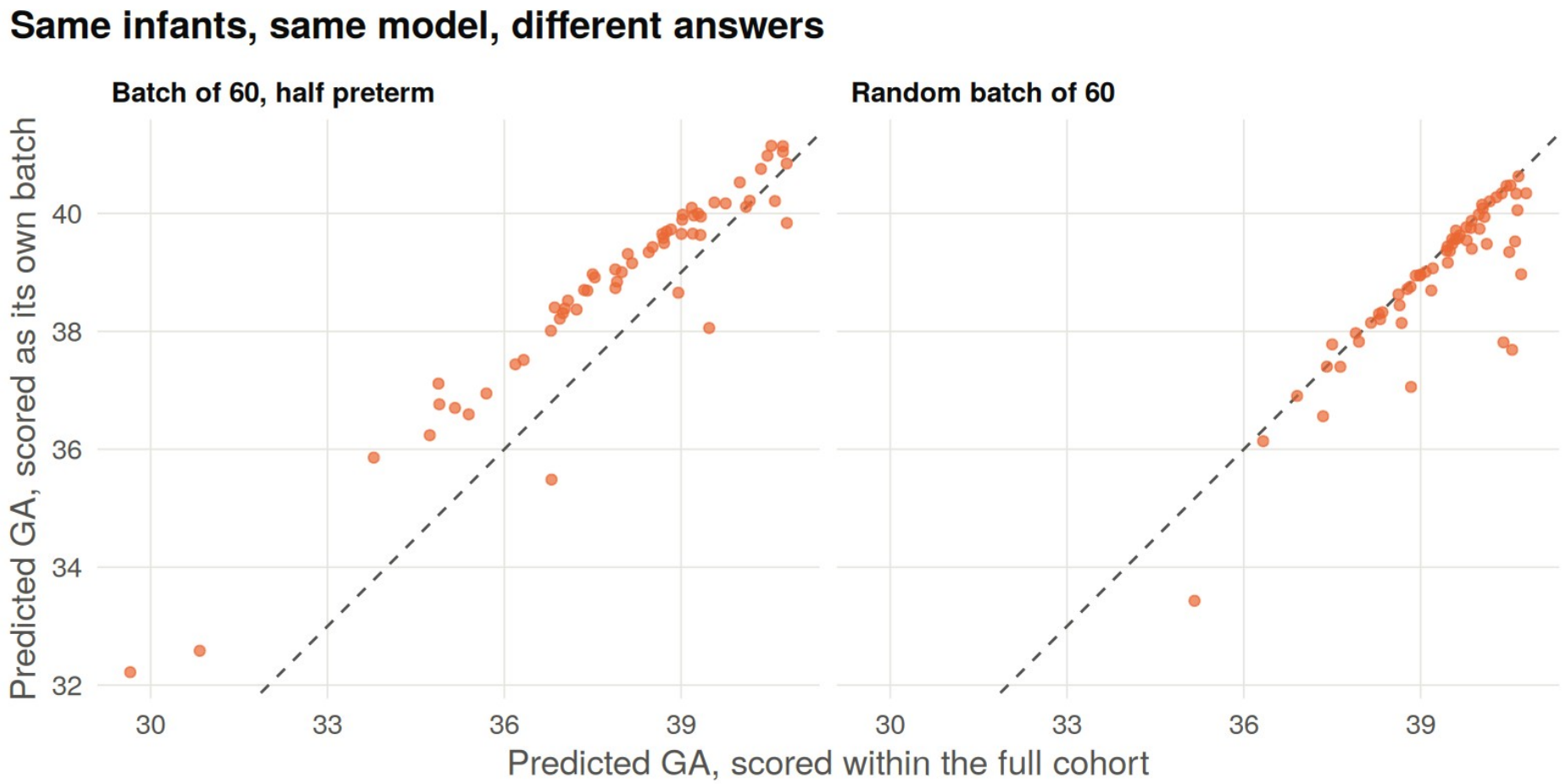
up to ½

how much the preterm birth rate can be underestimated by counting infants predicted < 37 weeks

- Regression to the mean again: predictions are pulled toward term
- Good individual predictions can still give a biased population count
- For surveillance, average the calibrated predicted probabilities instead of counting classifications

The intended use (Part 1) decides the right estimator.

Same infants, same model, different answers



Per-batch standardisation means an infant's prediction depends on who else is in the batch.

The fixes: freeze the pipeline, then update

Freeze every constant

- Learn the floor, winsorisation cap, centre, scale and spline knots once, in the reference data
- Store them with the coefficients and apply them unchanged
- Bonus: a covariate-shift table for every new cohort, in reference SD units

Update for the new setting

- Site A: well ranked but shifted
- Re-estimate the intercept from 750 local infants: the bias and most of the risk miscalibration disappear
- Adding a slope gains little here; the AUC doesn't change at all

Poor calibration in a new setting is often fixable by updating, if the pipeline is reproducible.

PART 7 · 6:40

What we'd do differently

A candid look back, then your questions.

Lessons from this case study

- The model is the whole pipeline. Learn every preprocessing constant in the training data only, store it, and test that predictions don't depend on the batch
- Store spline knots as numbers, and prefer restricted (natural) cubic splines, linear beyond the outer knots, for models that will travel
- Look at covariate shift before outcomes: sample type, laboratory, population
- If you standardise per site, do it on purpose, describe it as part of the model, and test its sensitivity
- Plan external validation sample sizes around events, not participants
- Justify class-imbalance weights, and report calibration
- Report the model as coded, following TRIPOD+AI

Take-home checklist

- What is the model for? The intended use decides which metrics matter.
- Report discrimination and calibration, never the AUC alone.
- Judge calibration conditional on the prediction.
- Check net benefit at thresholds that make sense.
- Internal validation estimates optimism; external validation tests transportability.
- Interrogate the reference standard before blaming the model.
- Poor calibration in a new setting is often fixable by updating.
- Handle class imbalance with thresholds, not distorted risks.
- Report with TRIPOD+AI; appraise with PROBAST+AI.

QUICK POLL · HANDS UP

Discussion: think of a prediction model you use or review. Which of today's checks would it fail?

A
Calibration never reported

B
No external validation

C
No net benefit / decision analysis

References and further reading

Hawken S, et al. Development and external validation of machine learning algorithms for postnatal gestational age estimation using clinical data and metabolomic markers. PLOS ONE 2023.

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Murphy MSQ, et al. External validation of postnatal gestational age estimation using newborn metabolic profiles in Matlab, Bangladesh. eLife.

Van Calster B, et al. Calibration: the Achilles heel of predictive analytics. BMC Med 2019.

van den Goorbergh R, et al. The harm of class imbalance corrections for risk prediction models. JAMIA 2022.

Vickers AJ, Elkin EB. Decision curve analysis. Med Decis Making 2006.

Riley RD, Archer L, et al. Evaluation of clinical prediction models (parts 1–3). BMJ 2024.

Collins GS, et al. TRIPOD+AI statement. BMJ 2024.

Data: Dryad, doi:10.5061/dryad.m37pvmd6b

Thank you

Questions and discussion

Companion R code and open data: rerun every example in this talk (demo.qmd, demo_real.qmd)

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